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Project Title:	Free Vortex Separator		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
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Zip Code:	20794	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/24/2011 with awarding of new grant, "Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator," grant NNX11AO76A; original end date was 5/31/2013 for NNX09AI35G. Information per PI and J. McQuillen/GRC (Ed., 3/8/2013) NOTE: End date is now 5/31/2013 (previously 5/31/2010), per NSSC information (jvp/editor 6/8/10)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Xiongjun (Dynaflow, Inc.) Loraine, Greg Ph.D. (Dynaflow, Inc.)		
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Task Description:

The limits on the amount of liquids and gases carried to space make it imperative to recycle and reuse them. On earth bubbles are relatively easily separable by buoyancy. In microgravity, other external forces such as a centrifugal force must be utilized to separate bubbles from liquids. Two categories of centrifugal force separators exist: one uses active rotation by mechanically spinning the tank. This is very efficient but requires shaft, bearings, and motor. In the second the tank is fixed, and the rotation is induced by eccentric injection of the mixture (Free Vortex Separator or FVS). This FVS has been proven very efficient for mid-range void fraction (50 -80% gas) with an efficiency approaching 100%. However, this FVS is not as efficient for lower void fraction, and there is a need of new research for low void fractions. We have developed a phase separator at DYNAFLOW based on a DYNASWIRL® swirl flow nozzle. It consists of two concentric cylinders, the fluid is introduced in the space between the two cylinders, and the swirl flow inside the inner cylinder is generated with tangential injection slots. With this configuration we can make the vortex circulation intense enough to generate cavitation with a very stable vaporous/gaseous core even at low flow rates. Operation with the high swirl near or slightly below cavitation inception in the vortex core guaranties removal of air even from mixtures with very low void fractions. This is achieved by forcing the bubbles to expand and collect in the vortex core. They are then removed through the central exit orifice. This configuration was tested successfully during a reduced-gravity flight. We are further developing and experimentally testing this configuration to characterize its gas-liquid separation performance and to improve its efficiency. We are also conducting numerical simulations to help optimize the design and ultimately build a compact system for space validation on the International Space Station (ISS) using the existing Fluid Integrated Rack facility.

[EDITOR's NOTE (3/8/2013): End date changed to 8/24/2011 with awarding of new grant, "Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator," grant NNX11AO76A; original end date was 5/31/2013 for NNX09AI35G. Information per PI and J. McQuillen/GRC. See that project for subsequent reporting.]

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

As a passive separator device, the DYNASWIRL® phase separator has the advantage of simplicity and dependability. It requires no moving mechanical parts and low power. Compared to other passive separator devices, which usually are not very efficient at low void fraction, the DYNASWIRL® is able to force low void fraction gas out of the liquid into the central core of the vortex by combining the effects of swirl, cavitation, and rectified gas diffusion. In addition to being an effective phase separator for space applications, it will also find wide applications in earth applications where fast separation of low void fraction gas is of concern.

Task Progress:

During this reporting period, we have continued our efforts to improve the performance of the prototype design of the DYNASWIRL® phase separator that demonstrated its performance and potential on a NASA reduced gravity flight test. We also conducted numerical simulations to help develop new design concepts and continued our efforts to develop a reliable in-line void fraction measurement device based on conductivity measurement.
A. Design of an Efficient Gas Extraction Outlet

The original prototype design used in the reduced gravity test had an orifice of fixed diameter for gas extraction. This orifice had to be smaller than the smallest gaseous core formed at the center of the DYNASWIRL® chamber at any operation condition in order to prevent liquid from entering the gas extraction line. However, since the core size varies with the operation void fraction and flow rate of the gas-liquid mixture, the higher the void fraction and flow rate, the larger the core size and vice versa. This fixed orifice size design thus puts a limit on the flux of gas being extracted from the core and reduces the overall efficiency of gas extraction especially for high void fraction mixtures. A gas extraction orifice with ideally variable size that matches the gas core size is therefore desired.

During this reporting period, we evaluated different design concepts for their feasibility to be used similarly to a gas extraction orifice of variable orifice size. A successful design concept we developed is based on the use of a hydrophobic membrane at a relatively large orifice size. The membrane has unique characteristics in that it only allows gas to pass through the membrane while blocking water as long as the pressure difference is less that a limit value for the membrane. We found out that regardless of the gas core size, the co-polymer membrane can provide the maximum area needed for gas extraction without allowing liquid to remix with the output. Feasibility tests have shown that the gas can be extracted efficiently through the membrane while the water can be blocked.

We are currently conducting extensive tests to select the optimal membrane pore size to be used in the phase separator.

Design of an Efficient Liquid Extraction Outlet

In the previous prototype designs, there was in the DYNASWIRL® chamber on the opposite end to where the gas extraction orifice was located a large liquid exit orifice. Since this exit orifice coincided with the gas core, a small fraction of the gas that was not extracted from the gas core also exited through the liquid exit orifice and re-mixed with the liquid. This remixing creates new gas-liquid mixture that has to be recirculated to be separated again.

To address this remixing issue, we have developed a new design that separates the liquid exit from the gas core utilizing annular openings for the liquid. With this design, a second gas extraction orifice with co-polymer membrane is used at the centerline inside the annular openings such that gas can be extracted from both end of the DYNASWIRL® chamber to enhance the gas extraction efficiency, while water is extracted from the annular region.

After a series of numerical simulation indicating the feasibility, a prototype was designed and constructed to verify the concept. In this prototype multiple small orifices whose total opening area is close to the opening area of the exit orifice in the original design are arranged in an annular area around the central orifice, around the gas extraction orifice. Our preliminary tests have shown that a stable gas core is formed at the center inside the liquid extraction opening boundary and no remixing of gas and liquid is observed at the liquid openings, indicating that this design can be used to eliminate remixing in the outlets. We are currently studying how to optimize the design.

B. Numerical Simulations

3D numerical simulations were conducted using our incompressible Navier-Stokes flow solver, 3DYNAFS-VIS©. This code includes bubble, cavities, and large free surface deformation effects and uses moving overset grids and dynamic grid generation schemes. 3DYNAFS-VIS© enables direct numerical solution in addition to RANS. The code is based on the artificial-compressibility method. It was used to characterize the flow field of the DYNASWIRL® phase separator.

Calculations were also performed on the geometry that included new design concepts under consideration to help design the new two-phase flow separator.

3DYNAFS-VIS© can be a powerful simulation tool to help optimize the annular opening design. 3DYNAFS-VIS© was used to study the influence on the flow and pressure field of the presence of an annular opening for liquid extraction. It showed that with proper choice of the size of the annular opening, it was possible to avoid remixing of gas and liquid. The simulation results were used to guide the prototype design successfully verified experimentally.

C. Instrumentation Improvement

To evaluate the performance of a phase separator, it is very important to measure properly the void fraction of the mixture. As shown in the flight test, the ABS ACOUSTIC BUBBLE SPECTROMETER® can be used to effectively measure the low void fraction from 10-3 and below. However, we still need an efficient and economic way to measure higher void fractions, such as in the inflow to the separator. In this reporting period, we continued our efforts to develop an in-house void fraction measurement device based on conductivity probes. The previously reported needle probe a was very fragile, therefore we used metal plates to construct new probes, Tests showed that the output from these probes gives a good correlation with the void fraction. A second set of electrodes in pure water was used as reference probe. The two probes were compared in order to see only the impact of bubbles and eliminate the drift in signals caused by the changes in electrical properties of the flow medium.

Currently we are developing electronic to improve the measurement accuracy.

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Bibliography Type: Description: (Last Updated: 09/04/2025)